

Precautionary Electrification*

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Abstract

I study the distributional effects of gas price volatility on households' energy adoption (extensive margin) and consumption (intensive margin). Two facts motivate the analysis. First, residential gas prices have historically been more volatile than electricity prices and are projected to remain so. Second, poorer households disproportionately rely on electricity, while richer households consume relatively more gas. I develop a heterogeneous-agent model of residential energy demand in which households with non-homothetic preferences choose between gas, whose price is stochastic, and electricity, whose price is stable. Calibrated to U.S. data, the model replicates observed patterns of fuel use and responses to price shocks. I find that higher volatility induces poorer households to shift toward electricity, while richer households remain exposed. These results highlight volatility as a driver of electrification and have substantial implications for the distribution of welfare, and energy policy.

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JEL Codes: .

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